CS 121 – Information Retrieval

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Assignment 3: Search Engine - Milestone 1

Index Analytics Report

Overview

For Milestone 1, our group built an inverted index using the DEV collection. The program reads the web pages, processes their HTML text, and stores the terms with the documents where they appear.

Index Statistics

Statistic	Result
Number of indexed documents	53,792
Number of unique terms	1,066,598
Merged index size	264,632.05 KB
Total index storage (including URL mapping)	269,146.07 KB
Partial index files created	11

How We Built and Checked the Index

To build the index, we started by reading the JSON files, assigning document IDs, and extracting text from the HTML pages. We then tokenized the text, converted the words to lowercase, and applied Porter stemming. The program also calculates term frequencies and keeps track of words that appear in titles, headings, and bold text. We used disk_indexer.py to build the inverted index in batches of 5,000 documents so we would not have to keep the entire index in memory. Our run created 11 partial index files, which were then merged into one final index. To check our results, we ran index_stats.py to count the indexed documents and unique terms and calculate the size of the saved index files. We processed 53,792 documents and found 1,066,598 unique terms. The final storage measurements are included in the table above.